

Bubble Sort

Definition

Bubble Sort is a simple comparison-based sorting algorithm. It repeatedly compares **adjacent elements** and swaps them if they are in the wrong order. After each pass, the largest unsorted element moves to the end.

Algorithm

1. Start from the first element.
2. Compare two adjacent elements.
3. If the first element is greater than the second, **swap** them.
4. Continue comparing adjacent elements until the end of the array.
5. Repeat the process for the remaining unsorted elements.
6. Stop when no swaps are required.

Output:

```
Enter number of elements: 5
Enter the elements:
4
8
9
7
6
Sorted Array:
4 6 7 8 9
Execution Time: 2.2172927856445312e-05 seconds

Time Complexity:
Best Case    :  $O(n^2)$ 
Average Case :  $O(n^2)$ 
Worst Case   :  $O(n^2)$ 
Space Complexity:  $O(1)$ 

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

- **Best Case:** $O(n)$ — when the array is already sorted and an optimized version detects no swaps.
- **Average Case:** $O(n^2)$
- **Worst Case:** $O(n^2)$

Space Complexity

- **Auxiliary Space:** $O(1)$
- Bubble Sort is an **in-place sorting algorithm**.